

Fluid motion and crystallization control enable air-processed fully screen-printed perovskite solar cells

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Fully screen-printed perovskite solar cells represent a promising solution for scalable manufacturing of perovskite photovoltaics. However, their development is hindered by inefficient vertical phase transformation within thick, multi-layered films, which results in incomplete infiltration, premature surface nucleation and residual stress. Here we report a fluid motion crystallization strategy employing a co-solvent system of ionic liquid methylammonium propionate and butyronitrile. The butyronitrile reduces the fluid motion resistance and disperses PbI_2 aggregates with optimized solvation structure, enabling rapid and deep infiltration of the perovskite precursor. This promotes a bottom-up, ordered phase transition before surface nucleation, effectively suppressing defect formation. The subsequent growth of a dense, interconnected perovskite nanocrystal network featuring an island-like surface morphology strengthens the interfacial contact with the carbon electrode. This intimate contact enhances the carrier transport and minimizes the recombination losses. Finally, air-processed, fully screen-printed perovskite solar cells yield a power conversion efficiency of 22.41% (certified 21.86%). Devices retain over 90.5% of their initial power conversion efficiency after 2,000 h under International Summit on Organic Photovoltaic Stability light-soaking-1 (ISOS-L-1) accelerated ageing conditions and show no degradation after 900 h of operation at 85 °C with 50% $\pm$ 10% relative humidity (ISOS-L-3 protocol).

The commercialization of perovskite photovoltaics is heavily dependent on the development of scalable and cost-effective manufacturing technologies^{1–4}. Fully printed perovskite solar cells (PSCs) are particularly promising owing to their compatibility with high-throughput, air-based processing, offering a scalable manufacturing route with reduced processing time and energy consumption. This technology integrates the unique properties of perovskite materials with established printing techniques, offering a low-cost solution for producing

high performance perovskite devices^{5,6}. In particular, screen-printing plays a critical role in the fabrication of perovskite thin films, enabling the deposition of high-quality perovskite thin films under ambient humidity conditions, with good morphology control and minimal material waste⁷. Fully screen-printed PSCs represent a significant advancement and have demonstrated significant potential for scalable and cost-effective production, where the entire device, ranging from the carrier transport layers to the perovskite active layer and carbon

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electrode layer, is fabricated through a single continuous printing process, enabling rapid manufacturing, reduced energy consumption, accelerated perovskite formation and excellent tolerance to ambient processing conditions⁸. Unlocking the full potential of this technology is essential for the further improvement device performance and advancing the commercialization of PSCs^{9,10}. However, precursor cluster formation, uncontrolled fluid motion and poorly regulated crystallization processes still limit device performance, resulting in the power conversion efficiency (PCE) of fully screen-printed PSCs remaining below 20%^{11,12}.

Here, we introduce an innovative solvent synergy strategy by incorporating butyronitrile (BN) into ionic liquid methylammonium propionate (MAPa) to control the perovskite fluid motion and crystallization process in fully screen-printed, carbon-electrode PSCs. The cooperative interaction between BN and MAPa tailors the solvation structure of the perovskite precursor, thereby improving the dispersion, enhancing the ink infiltration and enabling synchronous crystallization in screen-printed perovskite thin films. This strategy effectively lowers the phase-transition energy barrier within the mesoporous layer, enabling a complete phase transition in the deeper regions, which ultimately results in the formation of a high-quality, interconnected perovskite network within thick films. The high-density island-shaped morphology formed during this process also plays a crucial role in enhancing the interface between the perovskite and carbon electrode, promoting efficient charge transport. This strategy enables controlled fluid crystallization and low defect density in the screen-printed perovskite thin films, significantly improving the optoelectronic performance of fully screen-printed perovskite devices from 15.04% to 22.41%. Most importantly, under the ISOS-L-1 protocol, fully screen-printed PSCs maintain 90% of their initial efficiency after 2,000 h under maximum power point (MPP) continuous tracking, while maintaining 100% of the initial efficiency after 900 h of operation at 85 °C and 50% ± 10% relative humidity (RH) (ISOS-L-3 protocol), corresponding to an Arrhenius-equivalent T80 at 25 °C of approximately $(1.59\text{--}1.81) \times 10^5$ h.

Results and discussion

Fluid motion-controlled perovskite crystallization

The film formation process and crystallization behaviour of screen-printed perovskite thin films are significantly regulated by mesoporous materials, while the thickness of the mesoporous layer determines the choice of electrode in the devices (Supplementary Figs. 1 and 2). As shown in Fig. 1a, the fully screen-printed perovskite devices are fabricated layer by layer, enabling the controlled preparation of perovskite devices in ambient conditions. Figure 1b illustrates the structure of the mesoporous fully screen-printed perovskite devices, in which the crystallization process involves fluid infiltration, deep crystallization control and interactions at the carbon-electrode interface. BN (with a boiling point of 117 °C), as a polar co-solvent, improves the precursor solubility and maintains the chemical stability of the MAPa-based perovskite ink. Its exceptionally low viscosity and optimal boiling point provide a novel strategy for modulating the crystallization process of ionic liquid (MAPa, boiling point of 165 °C) perovskites with mesoporous materials (Supplementary Fig. 3)¹³. The crystalline quality of the screen-printed perovskite thin films prepared with different BN ratios was further evaluated, suggesting that a composition of 70% MAPa and 30% BN (denoted as MAPa + BN) may be beneficial for improving the crystallinity (Supplementary Fig. 4).

To clarify the role of BN in solvent regulation, the changes in viscosity of the perovskite ink and the particle size of the colloids were investigated. As shown in Fig. 1c and Supplementary Fig. 5, increasing the BN content suppresses the aggregate size in the perovskite ink and leads to a more uniform solute distribution. Meanwhile, the viscosity–temperature relationship and the rheological measurements in Fig. 1d consistently show that the MAPa + BN ink maintains a lower viscosity under both varying temperatures and printing-relevant shear

rates, which improves the ink flowability and contributes to more uniform film formation in the printing process¹⁴. The fluid motion behaviour of the perovskite ink was further investigated. As shown in Supplementary Figs. 6 and 7 and Supplementary Movie 1, the incorporation of BN enhances the wettability and droplet spreading of both the solvent and the perovskite ink on different substrates, thereby promoting the fluid mobility and facilitating deeper infiltration of the ink into the mesoporous framework. To elucidate the effect of BN on the chemical environment of the perovskite ink, Pb L-III edge X-ray absorption near-edge structure was conducted (Supplementary Figs. 8 and 9). The results indicate that BN incorporation slightly modifies the Pb–I interaction while preserving the Pb–O coordination, suggesting that the strong Pa–Pb coordination is preserved, thereby maintaining the chemical stability of the perovskite ink¹⁵. This conclusion is further supported by ¹H NMR revealing a reorganization of the local solvation environment (Supplementary Fig. 10a), ¹³C NMR and Fourier-transform infrared spectroscopy indicating preserved C=O coordination interactions within the precursor system (Supplementary Figs. 10b and 11) and UV–vis measurements confirming good long-term ink stability (Supplementary Fig. 12).

To further clarify the role of BN in the crystallization process, the evolution of the perovskite precursor ink and its crystallization behaviour within the mesoporous scaffold were systematically investigated. As shown in Fig. 1e, the intensity of the diffraction peak at $q = 10 \text{ nm}^{-1}$ is significantly enhanced in films prepared with the MAPa + BN perovskite ink, suggesting that the introduction of BN accelerates the crystallization process of the perovskite thin film, enabling more complete phase conversion within the mesoporous scaffold during annealing¹⁶. Figure 1f–h further elucidates the role of BN in the crystallization process within the mesoporous scaffold. During in situ X-ray diffraction measurements over a 900-s period, the control film exhibits a rapid increase in the intensity of the perovskite (110) diffraction peak, followed by a gradual decrease. By contrast, in the MAPa + BN film, a continuous increase in the (110) peak intensity is observed, indicating a broadened crystallization window and progressively improved crystallinity. Supplementary Fig. 13 shows that full width at half maximum broadening in the control film during prolonged annealing reflects increasing structural disorder, whereas continuous peak narrowing in MAPa + BN indicates progressively improved crystalline ordering. Supplementary Fig. 14 further reveals a stronger intermediate phase signal for MAPa + BN, accompanied by a peak shift from 9.54° to 10.27°, indicating the formation of a more compact intermediate structure. This compact intermediate structure is probably associated with the BN-regulated precursor organization, including smaller ionic liquid perovskite micellar aggregates and a thinner solvation shell, while the primary Pb–O coordination framework is preserved (Supplementary Fig. 15). Such precursor organization could facilitate more complete phase conversion and suppress defect formation during annealing, thereby contributing to improved crystalline ordering and device stability^{17–19}.

To further probe the BN-regulated crystallization kinetics, in situ photoluminescence (PL) and in situ absorption measurements were performed during film formation (Fig. 1i–l). A distinct three-stage evolution is observed for MAPa + BN. In stage I, a strong PL signal appears at the early stage, together with a distinct intermediate-state absorption feature, indicating that a dense Pa–Pb intermediate phase is rapidly formed. In stage II, the PL signal becomes stronger and narrower, while the absorption spectra continuously evolve towards the final band-edge feature, suggesting accelerated crystal growth and phase transition. In stage III, slight PL broadening is observed, which is attributed to continued solvent release and crystallization within the deeper mesoporous region. By contrast, the control sample exhibits a much weaker PL response and a less pronounced intermediate-state absorption feature, while the latter two stages are not clearly resolved, suggesting a less continuous and incomplete crystallization process.

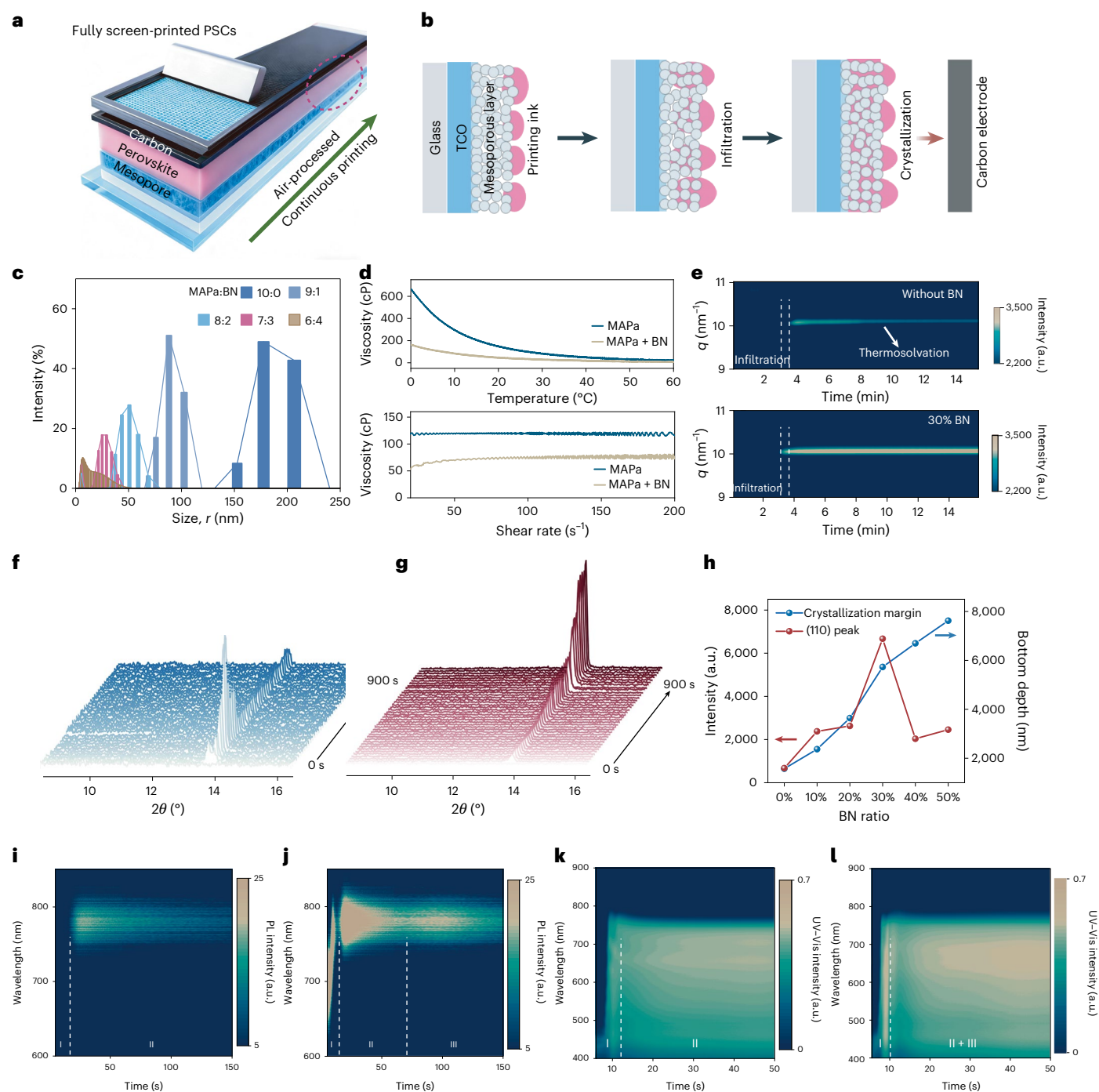


Fig. 1 | Infiltration and crystallization of perovskite printing inks. **a**, The preparation method for fully screen-printed PSCs. **b**, The device structure of fully screen-printed mesoporous carbon-electrode perovskite solar cells fabricated on a transparent conductive oxide (TCO) substrate. **c**, The dynamic light scattering particle size distribution for different ratios. **d**, The viscosity-temperature relationship and rheological measurements of MAPa and MAPa + BN solutions as functions of temperature (top) and shear rate (bottom). **e**, In situ GIWAXS of screen-printed perovskite thin films with (top) and without

(bottom) BN, where q denotes the magnitude of the scattering vector. **f,g**, In situ X-ray diffraction patterns of perovskite thin films without (**f**) and with (**g**) BN over 900 s. **h**, The (110) peak intensity and crystallization depth with different BN ratios. **i,j**, The time-resolved photoluminescence evolution during film formation in control (**i**) and target (**j**) samples. **k,l**, The time-resolved absorption evolution during film formation in control (**k**) and target (**l**) samples. All in situ thin film characterizations were performed on mesoporous substrates.

Theoretical analysis of perovskite fluid dynamics

To simulate the impact of BN on the fluid crystallization of perovskite thin films, we used Fluent for computational fluid dynamics simulations and Ab Initio Molecular Dynamics (AIMD) for atomic-level simulations. Figure 2a (left) shows a schematic of the mesoporous structure, where the green region represents areas of high fluid permeability (such as

air or mesoporous layer with large particles) and the blue region indicates areas of low fluid permeability (primarily composed of m-TiO₂ or m-ZrO₂). Fig. 2a (right), derived from Fluent simulations, visualizes the motion of perovskite particles as they move through these heterogeneous media, offering insight into the fluid dynamics that govern screen-printed perovskite crystallization process²⁰. Figure 2b

and Supplementary Fig. 16 show the distribution of perovskite particles in samples with and without BN. In the upper part of the graph, it is apparent that the particles predominantly aggregate at the top of the mesoporous structure, unable to penetrate to the bottom. This behaviour is attributed to the large particle size and restricted mobility, which hinder the diffusion and flow of particles throughout the mesoporous structure. In contrast, when BN is introduced, the mobility of the particles is significantly enhanced, enabling a more effective penetration throughout the interactive interface. This improvement in fluid dynamics facilitates uniform deposition and crystallization of the perovskite thin films, leading to enhanced structural integrity; the simulation details are provided in Supplementary Note 1. Furthermore, the regions of fluid crystallization can be defined according to the different crystallization stages (Fig. 2c), including the deposition phase (DP) region, high concentration (HC) region, medium concentration (MC) region, and low concentration (LC) region. The formation of these regions is strongly influenced by the solvent evaporation rate and spatial distribution across the mesoporous structure. An increased solvent evaporation rate typically facilitates the transition of Pb clusters from a dissociative state to a crystallized state, resulting in a larger crystallization zone and a reduction in unoccupied regions.

To further investigate the crystal evolution process, as shown in Supplementary Figs. 17 and 18, AIMD simulations provided insights into the temporal progression of the crystallization in samples with and without BN over a 3,000 fs timeframe. These simulations revealed how the solvent dynamics control the perovskite crystal formation and fluid interactions during crystallization. Figure 2d summarizes the effect of the solute-to-solvent concentration ratio on the phase transition of perovskite clusters at 3,000 fs. In the HC (10:5) region, Pb clusters are more dispersed, showing a similar distribution regardless of the presence of BN, which is attributed to the fact that the top perovskite ink remains in a more supersaturated state, promoting a stronger crystallization tendency. However, as the solvent-to-solute ratio decreases to MC (7:5) and LC (4:5), the Pb clusters in the BN-treated samples exhibit a more uniform dispersion. This demonstrates that BN effectively mitigates the aggregation of Pb^{2+} ions, enhancing the ordered arrangement of perovskite nanocrystals. The resulting dispersion state facilitates the involvement of organic ammonium salts in bonding, thereby promoting a more favourable phase transition. The results of visual molecular dynamics further confirmed the enhancing effect of BN on fluid crystallization (Supplementary Movie 2, representing the sample without BN, Supplementary Movie 3, representing the sample with BN, and Supplementary Fig. 19)²¹. The introduction of BN significantly accelerates the diffusion of the perovskite particles (Fig. 2e), promoting a more uniform distribution within the mesoporous structure. This enhancement in the particle mobility not only optimizes the diffusion pathways but also facilitates more homogeneous crystallization within the mesoporous layer, thus improving the structural stability of the perovskite thin films and reducing the formation of voids and defects. Moreover, the addition of BN significantly improves the spatial distribution of perovskite particles (Fig. 2f), promoting their transition towards a more ordered cubic phase. This improvement in particle alignment reduces the lattice distortion, thereby enhancing the crystallization quality and optoelectronic properties of the screen-printed perovskite thin films²².

Morphology regulation of screen-printed perovskite films

During ink spreading on the mesoporous scaffold, the fluid motion is governed by the initial flow and the Marangoni flow. At the early stage of thermal annealing, a relatively large initial flow is generated, promoting lateral spreading of the droplet over the mesoporous surface (Fig. 3a). As solvent infiltration proceeds, the fluid dynamics become dominated by competing Marangoni flows. The introduction of BN suppresses the inward flow (induced by temperature gradients) while preserving the outward flow (induced by concentration gradients),

enabling more uniform perovskite infiltration and crystallization within the mesoporous scaffold (Supplementary Figs. 20 and 21). As shown in Fig. 3b, Supplementary Figs. 22 and 23 and Supplementary Movies 4 and 5, in situ optical imaging reveals distinct phase transition behaviours in the two wet films. In the control film, large crystalline domains rapidly spread across the surface with obvious fluid redistribution, which hinders complete phase transition, whereas MAPa + BN shows smaller seeds and better footprint retention, indicating a more ordered and complete phase transition process. As shown in Fig. 3c and Supplementary Fig. 24, the PL signal in the mesoporous layer near the top surface already exhibits a noticeable difference, and the PL signal in the middle and bottom regions becomes even weaker and spatially less uniform for the control sample, indicating insufficient crystallization and greater local heterogeneity. By contrast, the target sample displays a markedly stronger and more spatially continuous PL intensity over extended areas in both regions, demonstrating that BN effectively improves the uniformity of the crystallized perovskite network. This conclusion is further supported by the luminance contrast observed in cross-sectional scanning electron microscopy (SEM) images acquired at a high accelerating voltage of 15 kV (Supplementary Fig. 25).

The crystallization results are further confirmed through high-resolution transmission electron microscopy (HR-TEM) and microstress analysis, which provide additional evidence for the marked reduction in microstress heterogeneity following the incorporation of 30% BN (Fig. 3d). Quantitatively, the mean absolute microstrain decreases from 0.0266 to 0.0086 after BN incorporation, indicating reduced local lattice distortion, which facilitates uniform and deep crystallization of perovskite nanocrystals within the mesoporous layers, as further supported by the fast Fourier transform patterns (Supplementary Fig. 26)²³. As shown in Fig. 3e, the control film shows a clear (110) peak shift from 14.03° to 14.11° with increasing tilt angle, indicating compressive residual strain according to Bragg's law, whereas this shift is markedly weakened after BN incorporation, indicating a significant reduction in residual stress, which implies that BN not only promotes deeper crystallization within the mesoporous layer but also contributes to a more uniform stress distribution, promoting the formation of well-ordered crystalline phases²⁴. As shown in Supplementary Fig. 27, two-dimensional grazing-incidence wide-angle X-ray scattering (2D-GIWAXS) analysis shows that, at 30% BN, the diffraction ring becomes sharp and continuous with maximal intensity, indicating enhanced crystallinity and a more complete phase transition within the mesoporous layers. Besides, the photoluminescence quantum yield increases from 0.827% in the control film to 5.903%, indicating suppressed non-radiative recombination (Supplementary Fig. 28). The fluid crystallization process of the perovskite ink plays a crucial role in determining the surface morphology of the deposited films. As shown in Supplementary Figs. 3f, g and 29, as the BN ratio increases from 0% to 30%, the island density on the mesoporous surface gradually decreases, while the grain size and surface coverage increase. However, when the BN content exceeds 30%, the film exhibits an overly aggregated morphology, which may limit uniform infiltration and hinder optimal crystallization. As shown in Fig. 3h, i, the high-density island morphology observed at 30% BN improves the interfacial contact between the perovskite and the loosely packed carbon electrode, which effectively prevents the accumulation of holes at the top interface. This structural evolution promotes the formation of an interlocking structure that could potentially enhance the open-circuit voltage (V_{oc}) by improving carrier transport.

Performance of fully screen-printed perovskite solar cells

Building on the research of perovskite fluid crystallization, we sequentially screen-printed the functional layers to fabricate fully screen-printed PSCs. The porous TiO_2 structure is uniformly filled with perovskite nanocrystals (Fig. 4a), while the sample without BN

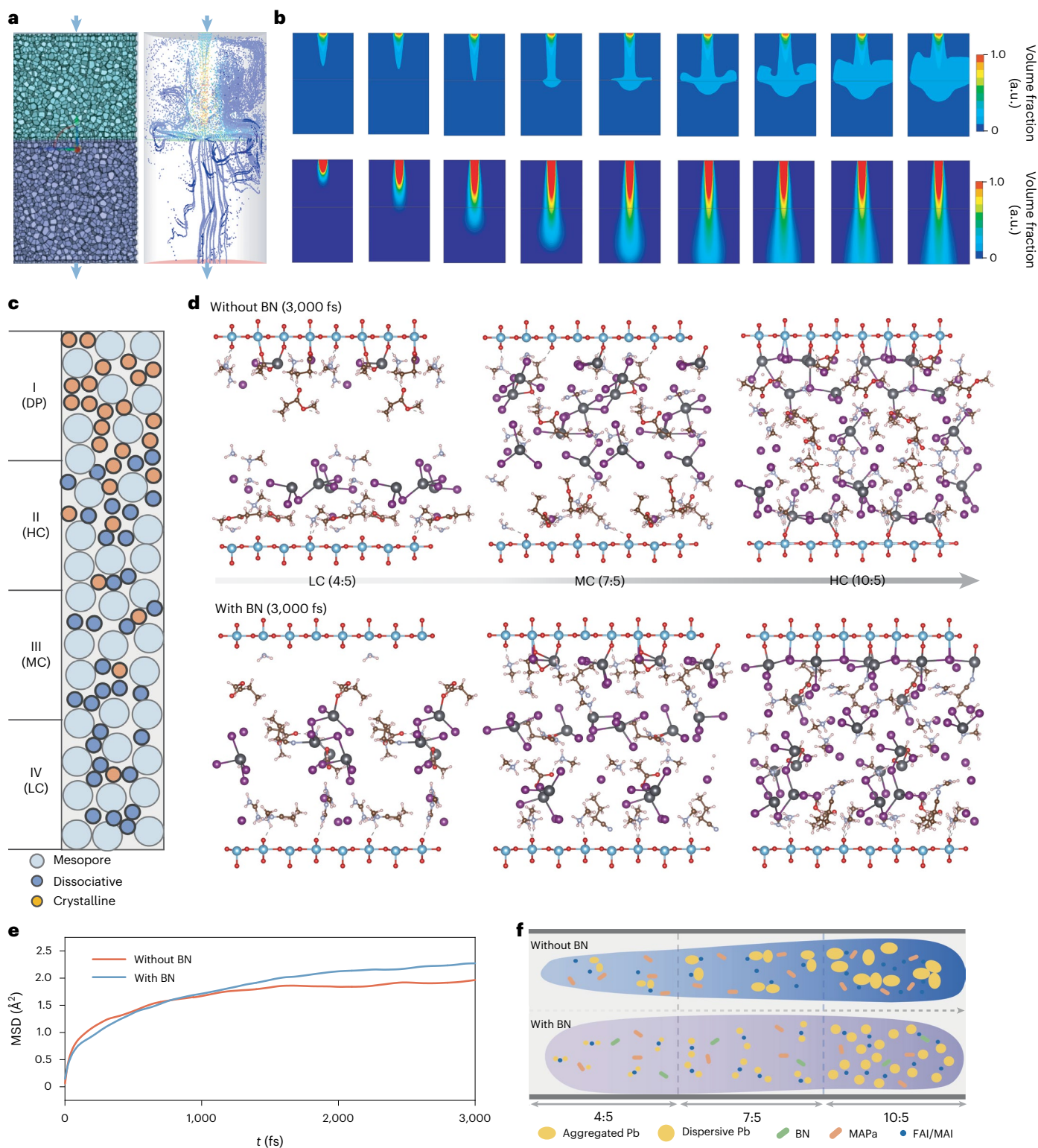


Fig. 2 | Perovskite fluid dynamic and crystallization simulation. **a**, A schematic of the mesoporous structure and fluid field simulation. The green and blue represent the mobile medium and mesoporous matrix, respectively. **b**, The fluid instantaneous kinestate of perovskite printing ink with (bottom) and without (top) BN. **c**, A schematic of vertical sampling segmentation according to the solute-to-solvent ratio, indicating a phase transition region, LC region (4:5), MC region (7:5) and HC region (10:5). **d**, AIMD results showing a comparison of

systems with and without BN at 3,000 fs across regions with different solute-to-solvent ratio. The cyan/red layers represent TiO_2 , while the purple/grey species represent Pb clusters. **e**, Mean square displacement (MSD) curves quantifying diffusion differences at various time points between systems with and without BN. **f**, A schematic of the particle distribution and phase transition along the depth direction.

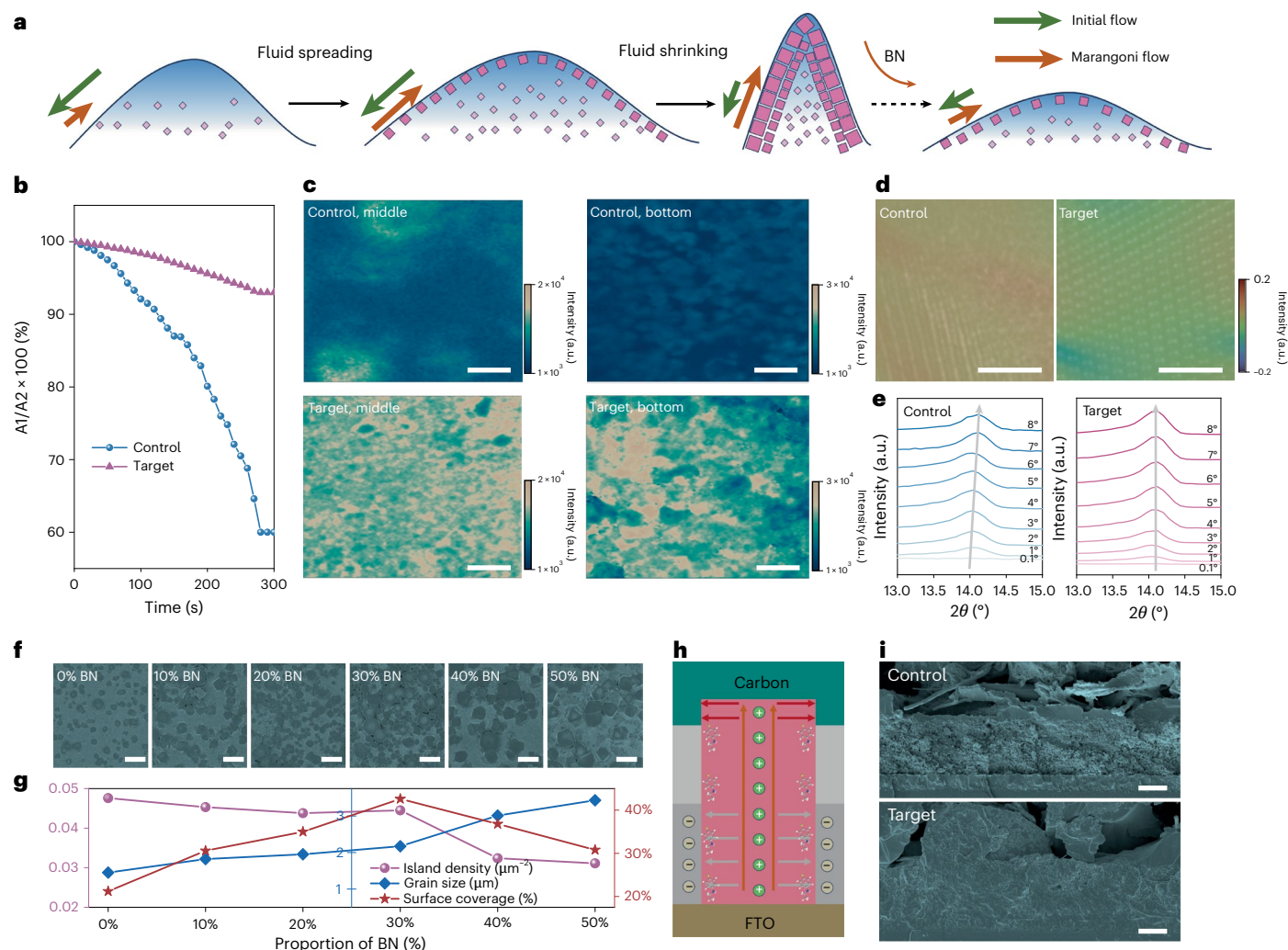


Fig. 3 | Crystallization characterization of screen-printed perovskite thin films. a, The fluid flow process of perovskite printing ink on the mesoporous surface during the thermal deposition process. **b**, The time-dependent projected footprint retention ratio of the control and target perovskite wet films during phase transition on a 50 °C hot stage from 0 to 300 s. A1 represents the area at a given time, and A2 is the initial area. **c**, Large-area PL mapping in the middle and bottom regions of the mesoporous scaffold. Scale bars, 50 μm . **d**, HR-TEM and crystal microstrain analyses images of screen-printed perovskite thin films with and without BN. Scale bars, 2 nm. **e**, GIXRD patterns of the perovskite thin films

with and without BN at different tilt angles. Grey arrows indicate the (110) peak shift. **f**, Surface SEM images of perovskite thin films with different proportions of BN under the same screen-printing conditions. Scale bars, 5 μm . **g**, The dependence of the average lateral island size, island density and surface coverage on the BN content. **h**, The device structure and carrier transport model of fully screen-printed PSCs. Green, pink, grey and brown represent carbon, perovskite, mesoporous scaffold and FTO, respectively. **i**, A cross-sectional SEM image of the interface between the island-shaped perovskite crystal and the carbon electrode. Scale bars, 1 μm .

exhibits numerous continuous nanoscale voids, suggesting a less efficient filling process and an uneven distribution of perovskite within the mesoporous layer (Supplementary Fig. 30). Moreover, the elements I and F exhibit a uniform vertical distribution within the mesoporous layers in the MAPa + BN sample, whereas the control sample shows a decreasing I signal along the depth, further confirming that BN promotes more effective perovskite infiltration and crystallization (Fig. 4b,c and Supplementary Fig. 31). The surface distribution of the element Pb at the bottom of the mesoporous layer further supports the role of BN in regulating fluid motion and crystallization (Supplementary Figs. 32 and 33). Additionally, the incorporation of BN significantly increases the surface potential, indicating suppressed non-radiative recombination and enhanced carrier extraction and transport (Supplementary Fig. 34). Meanwhile, more efficient carrier extraction at the bottom interface is achieved, along with the formation of a high-quality perovskite thin film (Supplementary Fig. 35). In addition, temperature-dependent capacitance–frequency (C–F) measurements reveal a weaker defect-related electrical response in

target devices. The activation energy decreases from 44.5 meV in the control sample to 27.1 meV in the target sample, indicating a reduced trap density (Supplementary Fig. 36).

On the basis of the optimization of film surface morphology and internal crystallization process, we fabricated high-performance fully screen-printed PSCs. As shown in Fig. 4d,e and Supplementary Figs. 37 and 38, the target devices fabricated in ambient air achieved the highest PCE of 22.41%, with a V_{oc} of 1.132 V, a short-circuit current density (J_{sc}) of 24.43 mA cm^{-2} and a fill factor (FF) of 0.810 (certified PCE of 21.86%)^{12,25–30}. In contrast, the control device exhibited a lower PCE of 15.04%, with an open-circuit voltage (V_{oc}) of 0.864 V, a J_{sc} of 23.90 mA cm^{-2} and a FF of 0.728. As shown in Supplementary Figs. 39–41, devices with 30% BN exhibit the best overall performance across different formamidinium/methylammonium/cesiums perovskite compositions and device architectures. Importantly, the scalability of this mechanism is supported by the fabrication of a fully screen-printed perovskite mini-module with an effective aperture area of 12.60 cm^2 . The module delivers a PCE of 20.13% with a V_{oc} of 6.72 V, a J_{sc} of 4.06 mA cm^{-2} and a FF

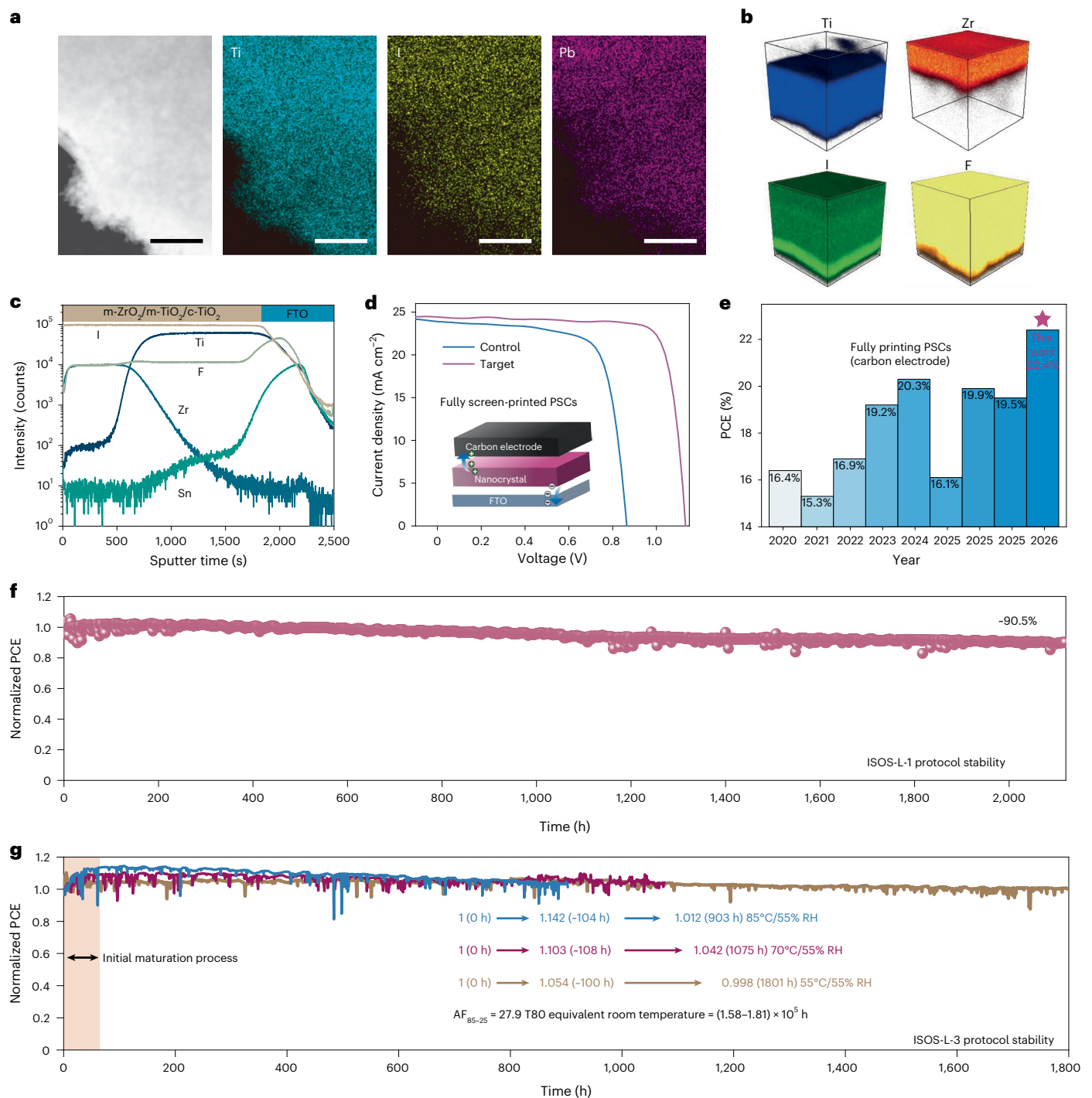


Fig. 4 | The device performance of fully screen-printed PSCs. a, The element distribution of the target screen-printed perovskite thin films as characterized by transmission electron microscopy-energy dispersive spectroscopy measurement. Scale bars, 200 nm. **b**, The three-dimensional spatial distribution of screen-printed perovskite thin films. **c**, Time-of-flight secondary ion mass spectrometry of the target film. **d**, $J-V$ curves of control and target fully screen-

printed PSCs. **e**, The evolution of the PCE for fully screen-printed PSCs with a carbon electrode reported over the years. **f**, The operational stability of fully screen-printed perovskite PSCs under continuous 1-sun illumination following the ISOS-L-1 protocol. **g**, MPP tracking of fully screen-printed perovskite solar cells under the ISOS-L-3 protocol.

of 0.738 (Supplementary Fig. 42). As shown in Supplementary Fig. 43, the integrated J_{sc} values from the external quantum efficiency measurement is 24.42 mA cm^{-2} , aligned with the $J-V$ curves. As shown in Supplementary Figs. 44–46, the target devices show lower dark current, reduced electron and hole trap densities and a smaller impedance arc. The electron trap density decreases from 4.38×10^{15} to $2.07 \times 10^{15} \text{ cm}^{-3}$, while the hole trap density decreases from 8.05×10^{15} to $5.41 \times 10^{15} \text{ cm}^{-3}$.

Meanwhile, the recombination resistance increases from 279 to 942Ω , confirming reduced leakage loss, suppressed trap-assisted recombination and more efficient charge extraction at the perovskite-carbon interface. The transient photocurrent measurements further show that the target device exhibits a weaker dependence of decay time on illumination intensity than the control device, indicating more stable and efficient charge extraction (Supplementary Fig. 47).

The enhanced device stability is further supported by ageing tests at 55 °C and 50% ± 10% RH, where the target film shows suppressed PbI_2 formation and continuous crystallinity improvement compared with the control (Supplementary Fig. 48). The long-term operational stability of perovskite photovoltaic devices is a critical indicator of their reliability and practical applicability. To rigorously evaluate their operational performance, we accelerated the ageing process by controlling multiple factors, including temperature, light exposure and humidity. Initially, we tested the light stability of the devices under the ISOS-L-1 protocol at 25 °C. As shown in Fig. 4f, the unencapsulated target device retained 90.5% of its initial efficiency after 2,000 h of MPP tracking under continuous light soaking (100 mW cm⁻²), demonstrating outstanding operational stability^{31,32}. Notably, as shown in Fig. 4g, the encapsulated target device showed an initial performance increase during the first 100 h of ISOS-L-3 testing at 85 °C and 50% ± 10% RH. This behaviour can be attributed to an initial maturation process involving trace moisture-assisted crystallization, slow residual solvent release and structural relaxation. We then combined single-temperature linear extrapolation with multi-temperature Arrhenius analysis based on MPP stability data measured at 55 °C, 70 °C and 85 °C. Specifically, for each temperature, we extracted the post-peak degradation behaviour from the MPP tracking curves and determined the corresponding characteristic decay time (Supplementary Note 2). The characteristic times derived from the three temperature points were then fitted using the Arrhenius relation, yielding an apparent degradation activation energy of $E_a = 0.510$ eV. On the basis of this experimentally derived E_a , we further calculated the corresponding acceleration factor (AF) from 85 °C to 25 °C, which is about 27.9, and obtained an Arrhenius-equivalent T80 at 25 °C of approximately $(1.59\text{--}1.81) \times 10^5$ h, demonstrating the excellent moisture and heat stability of the perovskite devices based on fully screen-printing technology^{33,34}.

Conclusion

We identified the role of BN in regulating the fluid dynamics and crystallization process of ionic liquid-based perovskite inks. BN significantly enhances the dispersion effect and infiltration process, yielding a more uniform filling of the mesoporous scaffold. These effects drive deeper crystallization and more complete phase transitions, producing screen-printed perovskite films with reduced defect densities. These improvements greatly enhance the morphology of screen-printed perovskite thin films and facilitate better carrier extraction at the carbon electrode interface. The air-processed fully screen-printed PSCs achieved a PCE of 22.41% (certified PCE of 21.86%). Furthermore, the fully screen-printed PSCs demonstrated excellent MPP stability under both the ISOS-L-1 and ISOS-L-3 protocols. This solvent strategy is compatible with ambient processing and continuous screen-printing technology, providing valuable insights for scaling up perovskite photovoltaics.

Online content

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Methods

Chemicals and reagents

All materials, including ethylamine solution (33 wt.% in absolute ethanol; Sigma-Aldrich), propionic acid (anhydrous, 99%; Sigma-Aldrich), dehydrated ethanol (anhydrous, 99%; Sigma-Aldrich), titanium isopropoxide (anhydrous 95%; Aladdin), acetylacetone (anhydrous, 99%; Aladdin), titania paste (m-TiO₂, 21 wt.%; GreatCell Solar), zirconia paste (m-ZrO₂, 21 wt.%; WonderSolar), formamidinium iodide (FAI, 99.9%; GreatCell Solar), methylammonium iodide (MAI, 99.9%; GreatCell Solar), low-temperature carbon paste (35 wt.%; WonderSolar), lead iodide (PbI₂, 98%; TCI) and 4-fluorophenethylamine (*p*-FPEAI), were used as received without further purification.

Theoretical simulations

CP2K was employed to carry out the theoretical calculations in the framework of density functional theory. CP2K/Quickstep employed two representations of the electron density: localized Gaussian and plane wave basis sets. For the Gaussian-based (localized) expansion of the Kohn–Sham orbitals, we used a library of contracted molecularly optimized valence double-zeta plus polarization basis sets, and the complementary plane wave basis set had a cutoff of 400 Rydberg for the electron density. The generalized gradient-corrected approximation of Perdew, Burke and Ernzerhof (PBE) was adopted to relax the geometric structures. The dispersion correction was applied in all calculations with the GrimmeD3 method. For sampling the structures of bulk solution, canonical ensemble (NVT) conditions were imposed by a Nosé–Hoover thermostat at a temperature of 300 K. The MD time step was set to 1.0 fs.

Synthesis of MAPa

Propionic acid (glacial, 17.6 ml) and methylamine (30 ml) were combined in a 500-ml round-bottom flask and stirred in an ice bath for 2 h. After the reaction mixture had been stirred at 0 °C for 2 h, the synthetic chemical MAPa was recovered by rotary evaporation at 60 °C for 1.5 h. The resulting primary product was then mixed with 30 ml of ethanol and further subjected to rotary evaporation at 85 °C for 1.5 h to obtain the final product.

Preparation of perovskite printing inks and screen-printed perovskite films

To obtain FA_{0.30}MA_{0.70}PbI₃ screen-printing perovskite ink, PbI₂ (295.5 mg), MAI (71.4 mg), FAI (33.1 mg) and *p*-FPEAI (15 mg) were mixed in MAPa (1 ml) or in solvents with different BN-to-MAPa ratios. The mixture was then stirred and heated at 60 °C for 8 h in a nitrogen-filled glovebox, producing perovskite printing ink. The screen-printed perovskite thin films were then fabricated in ambient air, using the ink with a printing speed of 2.5–40 cm s^{−1}, a printing gap of 0.3–0.7 mm, a printing pressure of 0.25–0.45 MPa, a substrate temperature of 20–60 °C and a mesh count of 150–500, and the printing perovskite thin films were subsequently annealed at 100 °C for 5 min to form the final perovskite thin films.

Fabrication of fully screen-printed perovskite devices

The compact TiO₂ layer, m-TiO₂ layer and m-ZrO₂ layer were sequentially screen-printed onto the substrate and sintered at 510 °C for 30 min in a muffle furnace. The perovskite ink was screen-printed onto the surface of ZrO₂ in ambient air. After 10–60 s of ink fluid infiltration, the wet film was annealed at 100 °C for 5 min. After that, the carbon electrode was screen-printed on the top of the perovskite thin film and annealed at 90 °C for 3 min.

Fabrication of a fully screen-printed perovskite mini-module

The fully screen-printed perovskite mini-module consisted of six sub-cells connected in series. P1, P2 and P3 were fabricated by laser scribing at different stages of the module fabrication process. P1 was performed

on a commercial fluorine-doped tin oxide (FTO) substrate before printing the functional layers to isolate the conductive FTO layer, with a scribe width of 30 μm. P2 was performed after the deposition of c-TiO₂, m-TiO₂, m-ZrO₂ and perovskite layers, removing these materials and stopping at the FTO layer, with a scribe width of 130 μm. P3 was performed after the deposition of c-TiO₂, m-TiO₂, m-ZrO₂, perovskite and carbon electrode layers, cutting through these layers and stopping at the FTO layer, with a scribe width of 100 μm. No intentional spacing was introduced between adjacent P1, P2 and P3 scribes.

Photovoltaic measurements

The devices were tested using a Keithley 2400 series source meter with a SS-F5-3A solar simulator (Enlitech, Taiwan) in a N₂ glovebox. The light intensity was calibrated to 100 mW cm^{−2} using a National Renewable Energy Laboratory (NREL)-standard monocrystalline Si cell. The performance parameters were obtained from the reverse-scan *J*–*V* curves of the photovoltaic cells under illumination with a scan rate of 20 mV s^{−1} and 1 ms delay time. The external quantum efficiency was measured by using an external quantum efficiency instrument (Enli Tech). The fully screen-printed devices were fabricated on 1.5 × 2.5 cm² substrates with a printed carbon electrode area of 1.0 cm². During *J*–*V* measurements, a metal mask with an aperture area of 0.049 cm² was used to define the illuminated area, and the photovoltaic parameters were calculated on the basis of this aperture area.

Characterization

Solution characterization. The viscosity of solvents was measured with a rotational rheometer (Anton-Paar MCR302). Dynamic light scattering measurements were carried out using a Malvern Zetasizer Nano instrument with a concentration of 200 mg ml^{−1} perovskite precursor solution. X-ray absorption fine structure spectra at the Pb L₃ edge (13,035 eV) of the precursor solutions were collected at the 1W1B beamline of the Beijing Synchrotron Radiation Facility. NMR spectra were recorded by using a Bruker Avance III HD500. X-ray absorption fine structure measurements at the Pb L₃ edge (13,035 eV) of the precursor solution were conducted at the 1W1B beamline of the Beijing Synchrotron Radiation Facility. The viscosity of the solvents and perovskite inks was measured by using a rotational rheometer (Anton Paar MCR 302). UV–vis spectra were measured with a Varian Cary5. Fourier-transform infrared spectroscopy was performed by using a Thermo-Fisher Nicolet 6700 system. Contact angle measurements were performed using a Dataphysics OCA20 goniometer. In situ optical microscopy was performed using a Leica DMI 8 microscope. Small-angle X-ray scattering measurements were carried out using an Anton Paar SAXSess MC2 instrument. In situ PL spectra were collected using a QE Pro spectrometer (ISPL-H1001; Nanjing Ouyi Optoelectronics Technology) with a 445-nm continuous-wave laser as the excitation source. In situ UV–vis absorption spectra were recorded using a PerkinElmer Lambda 950 spectrophotometer equipped with an integrating sphere.

Film and device characterization. The high-resolution transmission electron microscopy (HR-TEM) images were captured using a JEOL2100 from Japan. Absorption spectra were measured with a Shimadzu UV-2600 UV–vis spectrophotometer. SEM imaging was performed using a Hitachi S-4800 electron microscope with an electron beam accelerated at 5–15 kV. X-ray diffraction patterns were collected on a Bruker AXS D8 diffractometer using Cu Kα radiation. The atomic force microscopy and Kelvin probe atomic force microscopy measurements were carried out using a Bruker Dimension Icon atomic force microscope inside an Ar-filled glovebox. The time-resolved photoluminescence was recorded using an optronis optoscope streak camera with a temporal resolution of approximately 10 ps. The time-resolved photoluminescence was collected using an optronis optoscope streak camera which has an ultimate temporal resolution of approximately

10 ps, and the data were fitted with a biexponential decay model: $I(t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2)$, where $I(t)$ is the PL intensity at the time t , A_1 and A_2 are the amplitudes, and τ_1 and τ_2 are the decay lifetimes. Time-resolved photoluminescence spectra were measured with an Edinburgh FS5-TCSPC. The grazing-incidence wide-angle X-ray scattering data were acquired at the BL14B1 beamline of the Shanghai Synchrotron Radiation Facility. The space-charge-limited current measurements were conducted on electron-only and hole-only devices, with configurations of indium tin oxide/C60/perovskite/[6,6]-phenyl-C61-butyric acid methyl ester/Ag and ITO/Poly(3,4-ethylenedioxythiophene):Poly(styrenesulfonate)/perovskite/N,N'-Di(1-naphthyl)-N,N'-diphenyl-(1,1'-biphenyl)-4,4'-diamine/Ag, respectively. The J - V characteristics of SCLC measurements were recorded from 0 V to 5 V with a 0.02 V step size under dark conditions by using a Keithley 2400 source/meter unit. Elemental analysis, including Pb, I, F, Sn, Ti and Zr, was performed by plotting their intensities against sputter time to study their distribution and composition on the surface. UV-vis absorption spectra were recorded using a PerkinElmer Lambda 25 UV-vis spectrometer.

Operational stability measurements. The operational stability of the screen-printed device was measured at the maximum power point under ambient air conditions, using a white light-emitting diode lamp calibrated to simulate AM1.5G illumination (100 mW cm⁻²; Suzhou D&R Instrument Co., Ltd.). For ISOS-L-1 protocol stability measurements, unencapsulated devices were maintained at 20–25 °C under ambient temperature. For ISOS-L-3 protocol stability measurements, simply encapsulated (thermal sealing with butyl adhesive at 100 °C) fully screen-printed perovskite devices were subjected to ambient air at 50% ± 10% RH and different temperature (85 °C, 70 °C and 55 °C).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request. Source data are provided with this paper.

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Author contributions

Y.C. and W.H. conceived the idea and designed the experiments. Y.C. and Y.X. supervised the project. C.C. performed most of the device fabrication and characterization. Q.Y., Y.D., Y.Z. and A.C. assisted with the fabrication of perovskite films and related characterizations. C.C. and Y.C. contributed to data analysis and discussions. C.C. prepared the first draft of the manuscript. All authors discussed the results and provided feedback on the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	The area of tested solar cells is 0.049 cm ² . Explain why this information is not reported/not relevant.
Method used to determine the device area	☒ Yes ☐ No	Metal aperture mask. Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Current density-voltage (J-V) curves measured in both forward and reverse scan directions are provided in the independent certification report and in Supplementary Fig. 38.
Voltage scan conditions	☒ Yes ☐ No	Section "Photovoltaic measurements". Explain why this information is not reported/not relevant.
Test environment	☒ Yes ☐ No	Section "Photovoltaic measurements". Explain why this information is not reported/not relevant.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	Provide a description of the protocol. No preconditioning is required for our cell characterization.
Stability of the J-V characteristic	☒ Yes ☐ No	Section "Operational stability measurements", Fig. 4f, Fig. 4g. Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	Provide a description of hysteresis or any other unusual behaviour observed during the characterization. No unusual behavior was observed.
Related experimental data	☐ Yes ☒ No	Provide a description of the related experimental data. No unusual behavior was observed.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	We provide EQE measurement in Supplementary Fig. 43. Explain why this information is not reported/not relevant.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The integrated J _{sc} values are consistent with those from J-V measurements. Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell

☐ Yes
☒ No

Provide a description of the measurement conditions.

We didn't fabricate tandem solar cell.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

See details in Method.

Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

Our solar simulator was calibrated by Si reference cell.

Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☐ Yes
☒ No

Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements.

No spectral mismatch calculation was performed.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

An optical aperture mask (0.049 cm²) was used.

Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

Report the difference in the short-circuit current density values measured with the mask and aperture area.

No significant variations was observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

National Center of Inspection on Solar Photovoltaic Products Quality.

Explain why this information is not reported/not relevant.

A copy of any certificate(s)

☒ Yes
☐ No

Certification report provided in Supplementary Fig. 38.

Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

Fig.4d, Supplementary Fig. 38, 39, 40, 41and 42

Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes
☐ No

Fig.4d, Supplementary Fig. 38, 39, 40, 41and 42

Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

We provided stability data in Fig.4f and 4g.

Explain why this information is not reported/not relevant.